SHUBHAM SAPKAL

Pune, Maharashtra, India
Full Stack Software Engineer | Django REST Framework · Angular 18 · TypeScript · Python
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Summary

Full stack software engineer with 1+ years of production experience building **Django REST Framework** APIs and the **Angular 18** interfaces that consume them. Delivered a no-code YOLO model training platform that cut the time to train a custom model by 50%, a live HLS video analytics dashboard, and an offline document-intelligence desktop application for the HP AI Experience Center.

Technical Skills

- **Frontend:** Angular 18, TypeScript, RxJS, PrimeNG 17, HTML5, CSS3, responsive UI, HLS (.m3u8) video streaming
- **Backend:** Python, Django, Django REST Framework, RESTful API design, authentication and authorization (JWT, session), pagination, query optimization
- **Data and async:** MySQL, Firebase Realtime Database, Celery, RabbitMQ, WebSockets
- **DevOps:** Docker, Docker Compose, Nginx, Linux, Git, GitHub
- **Applied AI:** RAG, FAISS, Sentence Transformers, Ollama, YOLOv8
- **Languages:** Python, TypeScript, JavaScript (ES6+), SQL, Go

Professional Experience

Software Engineer | Augmented Transformations Pvt. Ltd.
May 2025 – Present | Pune, India

- Built a **no-code YOLO model training platform** with an Angular 18 and PrimeNG 17 interface where users upload, label and train in one place, replacing LabelImg annotation, custom scripts and manual server transfers; cut the time to train a custom model on 500 images from about 4 hours to 2 hours (**50%**).
- Built a video analytics dashboard and video wall in **Angular 18** and PrimeNG 17 that plays live **HLS (.m3u8)** feeds from multiple cameras in real time.
- Designed and built REST APIs in **Django REST Framework**, covering resource modeling, serializers, request validation, and authentication and authorization; owned the API contracts used by frontend teams and managed schema changes as requirements evolved.
- Reduced an API endpoint's response time from **12 seconds to 2 seconds** by replacing unbounded result sets with paginated responses and restructuring the underlying database queries.
- Developed an offline document-intelligence system for the **HP AI Experience Center** to analyze HP's internal documents (local LLM inference and vector search, no external API dependency), packaged as a desktop app for CPU, GPU and NPU.
- Consolidated duplicated logic into reusable Python/Django modules shared across multiple application modules.

Projects

SenitelAI – Traffic Violation Detection System | github.com/shubhamsapkal2912/SenitelAI
Tech: Angular 18 (TypeScript, RxJS), Django REST Framework, Python, RabbitMQ, Go, YOLOv8, Docker

- Full stack platform that detects overspeeding, riding without a helmet and red-light jumping from live RTSP camera feeds.
- Built the **Angular 18** dashboard with RxJS to show live violation feeds, evidence snapshots and filters from the **Django REST API**.
- Connected a Go ingestion service, RabbitMQ and Python YOLOv8 workers so stream I/O never blocks inference; deployed with Docker Compose behind Nginx.
- Handled **20 concurrent camera streams at 5 FPS** and detected about **90% of violations** across a 10-video test set.

CuratorAI – Offline RAG Document Intelligence System | github.com/shubhamsapkal2912/CuratorAI
Tech: Python, Celery, FAISS, Sentence Transformers, Ollama, Docker

- Fully local retrieval-augmented generation (RAG) system for uploaded documents, with no external API calls.
- Indexed documents of up to **97 pages** in the background with **Celery** workers, keeping API responses unblocked; built as independently replaceable services.

Education

Bachelor of Computer Applications (BCA) | CGPA: 8.54/10 ***Jul 2022 – Apr 2025***
MES Abasaheb Garware College of Science, Pune
Affiliated to Savitribai Phule Pune University